26900403 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEPES305 - SPORTS NUTRITION ESSENTIALS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. How can adequate protein intake contribute to athletic performance? [U] / [CO1]

   a). By supporting muscle protein synthesis

   b). By decreasing nutrient absorption

   c). By reducing muscle repair

   d). By preventing muscle contraction

2. Which nutrient provides a significant amount of energy during lower-intensity, long-duration exercise? [A] / [CO3]

   a). Fat b). Protein c). Vitamin C d). Calcium

3. When evaluating different psychological strategies for athlete weight management, which should be prioritized as most sustainable long-term? [An] / [CO3]

   a). Strategies focused solely on rapid appearance changes

b). Strategies with no long-term follow-up

c). Strategies that build resilience and healthy coping skills rather than quick-fix or appearance-focused approaches

d). Strategies that avoid any professional involvement

4. Reduced pre-exercise anxiety and improved confidence relate mainly to: [U] / [CO5]

a). Glycogen availability | b). GI comfort | c). Blood glucose stability | d). Psychological readiness

5. The main purpose of recovery nutrition is to: [A] / [CO2]

a). Restore glycogen, repair muscle and replace fluids

b). Cause dehydration

c). Reduce muscle repair

d). Eliminate carbohydrate

6. Which category of sports is most closely associated with pressure to make a specific weight class before competition? [K] / [CO3]

a). Casual jogging

b). Recreational swimming

c). Weight-category sports such as wrestling, boxing, and lightweight rowing

d). Long-distance road cycling

7. Which of these is a practical rehydration option? [A] / [CO4]

a). Plain black coffee | b). Fried snacks | c). Ice cream | d). Buttermilk (sambaram)

8. A small carbohydrate snack 30–60 minutes before exercise may provide approximately: [A] / [CO3]

a). 5–10 g carbohydrate | b). 30–50 g carbohydrate

c). 100–150 g carbohydrate | d). 200 g carbohydrate

9. Which of the following foods is a good source of calcium? [K] / [CO2]

a). Apple

b). Milk

c). Rice

d). Chicken

10. Adequate carbohydrate also has a protein-sparing effect because it: [A] / [CO2]

a). Reduces the use of amino acids for energy

b). Increases protein breakdown

c). Prevents protein absorption

d). Eliminates muscle repair

[1x10 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

11. Why are pre- and post-training meals important? [A] / [CO3]

12. Why is resistance training essential alongside a caloric surplus? [U] / [CO3]

13. Why should fluid intake be combined with carbohydrate and protein after exercise? [An] / [CO4]

14. What is the leucine threshold? [A] / [CO2]

15. Why can dehydration before competition impair performance? [U] / [CO4]

16. Explain the hydration plan recommended alongside pre-exercise meals [A] / [CO4, CO5]

17. Why should a healthy weight range be individualized for each athlete? [U] / [CO3]

[2x5 = 10]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

18. Explain the concept and purpose of carbohydrate loading? [A] / [CO3]

19. Discuss the role of sports nutrition in providing energy for physical activity and supporting muscle growth and repair. [U] / [CO1]

20. Explain why calorie-dense whole foods are preferred over processed foods for weight gain. [U] / [CO3]

21. Explain the "front-loading" and "shifting to liquid forms" strategies and how they work together across the hours before an event. [A] / [CO5]

22. Explain the general principles of hydration for athletes. [An] / [CO4]

23. Explain the general principles of hydration for athletes, including the balance between preventing dehydration and overhydration. [U] / [CO4]

[5x4 = 20]

Part D

Essay Type Questions

Answer any **2** questions

Each question carries **15** marks

24. Explain pre-exercise, during-exercise and post-exercise hydration strategies for endurance athletes? [A] / [CO3]

25. Discuss the nutritional principles underlying weight management in athletes, covering energy balance, energy density, macronutrient distribution, and the role of resistance training in preserving lean mass during a caloric deficit. [U] / [CO3]

26. Discuss the importance of proper nutrition in promoting health, development, tissue maintenance, and immune function. [K] / [CO1]

27. As a coach, design a checklist of "dos and don'ts" for athletes to avoid the common nutritional mistakes discussed in this unit, covering pre-exercise, during-exercise, and post-exercise phases. [A] / [CO5]

[15x2 = 30]